THE FUTURE OF DATA WAREHOUSING

database
TRENDS AND APPLICATIONS

Best Practices Series

SEVEN WAYS TO CAPITALIZE ON THE FUTURE OF DATA WAREHOUSING

Best Practices Series

Not too long ago, enterprise data warehouses represented the vanguard of analytics-driven organizations, as they gave decision makers their first taste of actionable information cut from various slices across business units. Financial planners had access to sales data, marketers had access to point-of-sale data, and operations teams had access to human resource data.

Now, a new generation of technologies—including the Hadoop framework, cloud, and in-memory processing—are propelling the staid data warehouse into bold new forms, running new types of workloads, supporting new approaches to data analytics, and housing data loads that dwarf the size and scope of traditional data warehouse configurations from just a few years back. Plus, today's new data warehouses are more powerful, versatile, and accessible from across the enterprise, and are designed to provide more well-rounded pictures of events and trends. They are no longer the large machines that sit in a corner of the data center; they are entire environments that are helping organizations complete their journeys to becoming analytics-driven cultures.

Never before has there been such an opportunity for enterprise data warehouses. Big data is only going to keep getting bigger, thanks to the proliferation of smartphones, devices, sensors, networks, and social enterprise information. Enterprise data warehouses will link data to key performance metrics and dashboards to aid decision support, serving as repositories of transformed information. At the same time, they will function as hubs to help manage the flow of big data and ensure it is ready for enterprise consumption. The data warehouse needs to be redefined and re-equipped to handle this emerging environment, and will continue to help lead the way for the data-driven decision making so critical in today's economy.

Here are seven ways to make the most of the new enterprise data warehouse:

1. LOOK TO THE CLOUD

Up until recently, building and managing an enterprise data warehouse involved a lot of time, resources, and money. Corporate IT staffs were charged with developing the skills and resources to build and manage the hardware, software, and systems necessary to make enterprise data warehousing a reality. Getting to this point may have already meant steep investments of money, plus months of project work to prepare data for the warehouse—not to mention the programming and administration work needed to make sure everything is running smoothly.

Cloud-based data warehouses may change this requirement. Instead of the months, or even years, involved in buying, installing, and learning the ins and outs of an enterprise data warehouse, managers can deploy a cloud-based data infrastructure in a matter of minutes, at least in theory. Little or no knowledge of data warehousing technology is required when using configurations from the cloud. The use of cloud for data warehousing may even become a necessity with organizations increasingly moving data assets to cloud providers. A number of vendors are releasing cloud-based data warehousing solutions—either as traditional, well-known databases accessible via cloud services, or new types of cloud-friendly data environments.

2. ADAPT AND INTEGRATE WITH HADOOP AND DATA LAKE ENVIRONMENTS

While in times gone by the enterprise data warehouse resided in a single system, the data warehouse that is evolving may bring together multiple analytics platforms. Along with varying data formats, this includes multiple reporting and programming languages as well. This serves as a complement to implementations of data lakes built in the Hadoop framework. Hadoop is designed to cost-effectively process even the largest datasets, employing commodity-based clusters across HDFS, its distributed file system, and its MapReduce processing capability. In this light, Hadoop can be the optimum environment to run

NEVER BEFORE HAS THERE BEEN SUCH AN OPPORTUNITY FOR ENTERPRISE DATA WAREHOUSES.

alongside traditional data warehouses, taking on growing workloads that may be too expensive or cumbersome to run in the warehouse. The movement toward data lakes—repositories that hold data in native formats until called by applications—also promises greater efficiencies. Enterprises will need to adapt their data warehouses to support Hadoop and accompanying data lake approaches, serving as ETL engines to move data to selected applications.

3. DECENTRALIZE

In their earliest incarnations, enterprise data warehouses were contained within hardware configurations, but in recent years, they have also been based on more distributed or logical configurations. A central repository—intended to store and manage physical data stores—is no longer essential to have a highly functioning enterprise data warehouse environment. The enterprise data warehouse is evolving to a far-reaching platform that can handle multiple types of data and multiple analytic workloads. Many organizations, in fact, are looking at "data as a service," or even "analytics as a service" approaches to serving decision makers with actionable enterprise information and insights.

4. AUTOMATE

Automated tools and environments are picking up the often arduous tasks involved in transforming and managing data going into enterprise data warehouse environments. At one point, for example, ETL scripts to capture data were manually coded. The manual processes involved in changing configurations, revising requirements, extracting data—which often took a long time to complete—can now be delivered much faster with automated tools. The key to such automation is the ability to speed up implementations, and thus, decision making. Some vendors now offer automation tools to accelerate data warehouse operations, and thus, enable faster implementations, faster and more cost-effective development, and accelerated decision making. Ultimately, enterprise data warehouse functions will be running in the background, behind the scenes, in an almost automatic fashion and invisible to users.

5. LIGHTEN UP

Increasingly, enterprise data warehouses are evolving from hardwired megaservers that need to be maintained in data centers to something more flexible, adaptable, and agile that changes as the business changes. The enterprise data warehouse will be far more distributed, with all or some pieces running in the cloud. It will increasingly exist as a service—or set of services—versus technology that needs to be maintained and upgraded. An important element will be the design of front-end interfaces that provide access to those back-end services, oblivious to where they reside, or how much they change. Those user interfaces are most likely to be either a browser or a mobile device app, which amount to extremely lightweight clients that require little or no upkeep on the part of enterprises. With all the changes taking place with technology these days, enterprises cannot get locked into arrangements with hardware and software vendors. Standards and configurations should be based on open protocols and offerings, to enable enterprises to easily make the shift between providers when needed.

6. CONSOLIDATE

There are a range of technology solutions being used to reduce the footprint and increase the capabilities of data centers while also boosting the value and efficiency of enterprise data warehouses. Strategies include virtualization, public and private cloud, data compression, and ever-expanding storage and CPU capacity.

7. OPERATIONALIZE FOR REAL TIME

Enterprise data warehouses are evolving from back-end, historical data repositories to real-time decision-support engines. As enterprises move to real-time analytics and capabilities, they require data warehouse capabilities that also represent the freshest information. The challenge is that data warehouse information has been based on ETL processes, meaning it takes time to bring data to decision-making applications.

Data warehouses' evolution to support the real-time enterprise is based on the development and deployment of data services that merge data from multiple sources, including real-time feeds as well as ETL-based historical data. This is particularly key with systems that reach out to customers, covering areas such as sales tracking, upselling, cross-selling, customer service, fraud analysis, risk management, and sentiment analysis. These functions would go into a real-time, operational data warehouse that is able to integrate data from live, ongoing engagements or transactions along with historical data in order to provide on-the-spot context to decision makers and applications.

—Joe McKendrick

The Enterprise Data Warehouse in the Age of Big Data

Life used to be simple. For a long time, Enterprise Data Warehouses were the central repositories of integrated data from disparate sources. They would store current and historical data and allow senior management to build reports to make critical business decisions. Companies would spend tremendous amounts of money and time building and maintaining these data warehouses. They were the main source of truth for the enterprise.

About ten years ago, humankind started generating data at an exponential rate and in unprecedented ways. In fact, 90% of the data in the world today has been created in the last two years. The Internet was originally responsible for most of this new data. However, today, sensors, machines and computers are now generating quintillion bytes of data every day. Even more impressive: about 80% of the information created and used by an enterprise is unstructured data. This figure is growing at twice the rate of structured data.

Many questions arise. Among them: How do you unlock the business value contained in this data? Can your organization store and analyze the massive quantities of data produced daily? Are you able to look at large volumes of historical data to identify patterns and derive new insights? What do you do with your unstructured data? Can you match your internal data with external sources to derive further intelligence? Are you able to do all of this in real time?

THE SHIFT

The promise is big, and the opportunity is real. Yet, for most companies around the world, big data has brought more questions than answers. Today, the explosion of new data types puts organizations under increased pressure to modernize their current data infrastructure with new platforms. As a result, many companies are looking to free their Enterprise Data Warehouse from data and workloads of unknown value and, instead, leverage that same data and workloads for operations that help them differentiate their business and make them more competitive in the marketplace.

Apache Hadoop has enabled businesses of all sizes to utilize big data cost-effectively and re-allocate the limited space in the Enterprise Data Warehouse only for that data that needs to travel first class. Hadoop is an open-source software framework that allows for the distributed processing of large data sets across clusters of hardware using simple programming models. It is designed for high availability and fault tolerance and can scale from a single server to thousands of machines. But Hadoop alone is just part of the solution.

Cisco, Hortonworks, and Informatica have teamed up to help achieve massive scalability. The Cisco UCS fabric-based architecture uniquely integrates server, network, and storage. The Hortonworks Data Platform sits on top to deliver enterprise Hadoop at scale. Cisco UCS, Hortonworks Data Platform, and Informatica Big Data Edition accelerate the journey to a Modern Data Architecture by enabling organizations to store and manage data more efficiently. By offloading ETL/ELT workloads and archiving infrequently used data online into Hadoop, organizations can accommodate the growth in data volume, variety, and velocity. The separate EDW and Hadoop environments can be federated using Cisco Data Virtualization, allowing business intelligence and analytics apps to take advantage of data in both data sources for richer insight.

Data warehousing systems can be optimized for higher value utilization by offloading Extract-Load-Transform (ELT) and Extract-Transform-Load (ETL) workloads, capturing vast amounts of streaming data and archiving infrequently accessed data onto more economical and scalable platforms like Apache Hadoop. The Cisco UCS fabric-based architecture uniquely integrates server, network, and storage. The Hortonworks Data Platform sits on top to deliver enterprise Hadoop at scale.

Even as the volume and variety of their data rapidly expand, enterprises can be confident they will be able to store and manage their data flow. The Cisco UCS fabric-based architecture uniquely integrates server, network, and storage. This highly efficient infrastructure lets businesses manage up to 10,000 UCS servers as if they were a single pool of resources so they can support the largest data clusters. And, as enterprise's big data deployments grow in size, the use of Cisco Unified Fabric technology significantly lowers costs by reducing the number of switches and cables they will require.

COMMON PAIN POINTS

Traditional enterprise data infrastructure is not optimized to meet modern data demands. The capacity of traditional enterprise data warehouses is often quickly consumed by these increased data volumes, leading to performance degradation and costly upgrades. As a consequence, data services teams often hedge against this performance degradation and expense by forcing information consumers to sample or aggregate data, thus limiting the quality and comprehensiveness of analytics. Limits on the fidelity of data directly diminishes the impact that information consumers, like data scientists or business analysts, can have on the business.

As part of data infrastructure modernization initiatives, leading organizations are jumping on the trend of data warehouse optimization. Data warehouse optimization enables organizations to offload infrequently used data and non-analytical Extract-Load-Transform (ELT) and Extract-Transform-Load (ETL) workloads from the enterprise data warehouse onto more economical, scalable, and modern platforms like Apache Hadoop, so that the data warehouse can be focused on supporting business operations with high performance analytics. The result is a more balanced data infrastructure portfolio with more scalable processes. ELT and ETL are executed on Apache Hadoop, while more performance-intense processes, like reporting, are executed on the data warehouse. The distributed data environments can be federated using Cisco Data Virtualization to provide business intelligence and analytics with a single point of access to all data.

But despite the obvious economic and business benefits of data warehouse optimization, organizations hesitate to embark on these projects for two reasons. The first reason is that organizations simply do not have the resources to manually develop data preparation components, like connectors, parsers, and transformations to support data infrastructure modernization initiatives. Furthermore, organizations lack the resources to tackle new hardware procurement, installation, and configuration projects. Data services teams typically aspire to quickly develop analytical solutions for their information consumers rather than be caught up in lengthy modernization initiatives. The second reason is that organizations often have already built up internal expertise and skills in developing ELT and ETL processes using automation on the data warehouse. Companies are reluctant to adopt new data platforms without the simplicity of re-using their existing expertise and leveraging proven hardware configurations to maximize success. These obstacles around time-to-market and risk inhibit the realization of the benefits of data warehouse optimization.

CISCO, INFORMATICA, AND HORTONWORKS—THE VISION

As leaders in their respective fields, Cisco, Hortonworks, and Informatica have partnered together to offer a comprehensive Data Warehouse Optimization solution for accelerating and simplifying the journey to a Modern Data Architecture. Data warehouse optimization projects are streamlined through the use of pre-built data preparation components and pre-built hardware systems, while project risk is lowered by re-using existing skills and leveraging proven hardware configurations. Using the joint solution, organizations can quickly and easily modernize their data infrastructure with Apache Hadoop to store data more efficiently and process data faster.

The economic foundation for data warehouse optimization is that Apache Hadoop and Hortonworks offers the industry's only completely open Hadoop platform. Through an open approach, Hortonworks Data Platform not only delivers the compelling economics and scalability of Hadoop, but also ensures supportability with the backing of a large open source community.

Data warehouse projects are accelerated and simplified by Informatica through the Informatica Big Data Edition. Informatica Big Data Edition ingests all types of data in batch and real-time using hundreds of pre-built parsers, connectors, and transformations. Informatica developers can then rapidly profile, parse, integrate, cleanse, and refine the data using a simple visual development environment and reusable business rules. With over 100,000 Informatica-skilled developers, organizations can leverage existing skills without any risk to ongoing maintainability.

Cisco Data Virtualization federates disparate data (Hadoop clusters and DWs), abstracts and simplifies complex data, and delivers the results as data services or relational views (that is, logical business views) to consuming applications such as business intelligence and analytics tools and other information dashboards. With advanced query optimization technology, it delivers extremely high performance.

Data warehouse projects are accelerated and simplified by Cisco through the Cisco UCS Common Platform Architecture for Big Data. Cisco UCS delivers a pre-built, high performance, and highly scalable hardware platform, combining compute, networking, storage, and unified management for meeting the demands of big data. The Cisco UCS solution accelerates deployments by delivering a pre-configured appliance for data warehouse optimization projects, while reducing implementation risk by leveraging a pre-tested solution.

CONCLUSION

As leaders in their respective fields, Cisco, Hortonworks, and Informatica are enabling organizations to modernize their data infrastructure quickly and easily. The combination of best-of-breed software and hardware combined with world-class professional services offerings from all three companies provides organizations and trusted approach for tackling data warehouse optimization projects. Using Cisco, Hortonworks, and Informatica, companies like yours can turn more data into actionable information that fuels competitive advantage.

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snowflake

Reinventing the Data Warehouse

THE EXPLOSION and rapid evolution in data and analytics is forcing businesses to reexamine their data warehousing solutions to determine what is needed to manage and leverage their data to strategic advantage.

THE CHANGING NATURE OF DATA

It used to be the case that most data you wanted to analyze came from sources in your data center: transactional systems, enterprise resource planning (ERP) applications, customer relationship management (CRM) applications, and the like. The structure, volume, and rate of the data were all fairly predictable and well known.

Today a significant and growing share of data—application logs, web applications, mobile devices, and social media—comes from outside your data center, even outside your control. That data constantly evolves and as a result frequently uses newer, more flexible data formats such as JSON and Avro. That increases demands on both the systems themselves and on the people who manage and use them.

THE NEED FOR A NEW DATA WAREHOUSE

As companies seek solutions that can accommodate the changes in data and how it is used, they're frustrated to find that most current solutions simply can't keep up. For example, traditional data warehousing solutions were designed for structured, predictable data. Yet companies today also see opportunity to gain insight from machine-generated data, i.e., data generated by web servers, cloud applications, sensors and more. At the same time, newer "big data" platforms like Hadoop are not data warehouses and require specialized expertise that is in very short supply.

As a result, companies are demanding a solution that can work with all business data, handle all their reporting and analytics workloads, and support all of their users in a single solution.

That requires a system that can natively handle both structured and semi-structured data such as weblogs, XML and JSON; scale up and down on the fly to support any number and intensity of workloads; make that data accessible through SQL so that they can leverage their current skills and tools; and do so at a fraction of the cost of traditional systems.

SNOWFLAKE TAKES A FRESH APPROACH

Snowflake was founded by a team of veteran experts in database technology who saw the opportunity to reinvent the data warehouse to meet these needs. They shared a vision to leverage new and disruptive technology including cloud infrastructure to bring together all users, data and workloads in a single data warehouse without compromising performance, flexibility or ease of use.

The result: the Snowflake Elastic Data Warehouse, a new SQL data warehouse built from the ground up for the cloud to deliver the power of data warehousing, the flexibility of big data platforms and the elasticity of the cloud—at 90% less than on-premises data warehouses.

What makes Snowflake's solution unique?

- **Data warehousing as a service.** Snowflake eliminates the pains associated with managing and tuning a database. That enables self-service access to data so that analysts can focus on getting value from data rather than on managing hardware and software.
- **Multidimensional elasticity.** Snowflake's elastic scaling technology makes it possible to independently scale users, data and workloads, delivering optimal performance at any scale. Run any number of workloads concurrently, including both loading and querying, because every user and workload can have exactly the resources needed, without contention.
- **Single service for all business data.** Snowflake brings native storage of semi-structured data into a relational database that understands and fully optimizes querying of that data. Query structured and semi-structured data in a single system.

Importantly, because Snowflake was built for standard SQL, it delivers these capabilities in a way that takes advantage of the tools and skills companies already have.

THE CLOUD AS THE FOUNDATION

Cloud infrastructure is the perfect platform for constructing the ideal data warehouse. It delivers near-unlimited resources, on demand, in minutes, and you only pay for what you use.

A data warehouse built for the cloud eliminates the obstacles and pains of deploying and managing infrastructure so that companies can focus on using their data rather than on dealing with infrastructure. In addition, the cloud is the natural integration point for data because a rapidly increasing share of the data comes from applications and systems outside the datacenter—cloud applications like Salesforce, web applications, mobile devices, sensors, and more.

A NEW STANDARD IN DATA WAREHOUSING

The Snowflake Elastic Data Warehouse brings sorely-needed innovation to data warehousing. Its unique architecture and capabilities offer a highly efficient, cost-effective, and easily manageable data warehouse service designed for today's business needs.

SNOWFLAKE
www.snowflake.net @snowflakeDB

MarkLogic

MarkLogic New Generation Data Warehouse Solutions

Process all data types with cloud-like scalability, reduce your time to value and your ongoing costs—without sacrificing data governance

For decades, enterprises have relied on an Enterprise Data Warehouse model as the foundation for cross-line-of-business decision support. These costly multi-year development initiatives have netted results that often fall well short of expectations. In today's Big Data world, organizations need new approaches—to deliver results more quickly, across a wider set of data assets.

MarkLogic delivers heterogeneous data integration and management, securely, with consistency—and does it at scale—to support both operational and analytic applications on the same platform. As the only Enterprise NoSQL database platform, MarkLogic combines the agility of NoSQL platform flexibility with powerful built-in search and semantics—and trusted enterprise-grade reliability, data integrity, and security. It is a new generation database that is built with a flexible data model to store, manage, and search today's data, without sacrificing any of the data resiliency and consistency features of last-generation relational databases. With these capabilities, MarkLogic is ideally suited for making heterogeneous data integration simpler and faster and for doing dynamic content delivery at massive scale.

OPERATIONAL DATA WAREHOUSE (ODW)

The ODW provides organizations with a single source of truth, scales to hundreds of nodes, and supports mixed workloads in a fully ACID-compliant implementation. Like other data warehouses, the ODW integrates data from operational systems for purposes of analytics. But, with a MarkLogic ODW, enterprises can:

- Handle structured, unstructured, and semantic data in a single platform
- Refresh the data store based on changing events—in real time
- Support user-driven data enrichment, letting business analysts "talk back" to the data warehouse

LOGICAL DATA WAREHOUSE (LDW)

The LDW is an umbrella architecture strategy designed to address the shortcomings of traditional Enterprise Data Warehouses. The LDW architecture includes:

- A repository to support data consolidation
- Virtualization capabilities to support federation of queries across data silos
- Distributed batch processing, e.g., to include use of Hadoop

MarkLogic expands upon the vision of the LDW, providing all LDW capabilities in a single technology stack— handling a variety of data types at massive scale in mission-critical environments. With MarkLogic, enterprises get fewer moving parts and quicker time-to-value—delivering results in weeks instead of years.

SOLVE COMPLEX DATA CHALLENGES

MarkLogic is optimized for structured and unstructured data—allowing you to store, manage, query and search across JSON, XML, RDF, geospatial data, text, and large binaries. This means faster time to value and an improved ability to handle change.

Here are some additional benefits to implementing an Enterprise NoSQL solution from MarkLogic:

Any Format—Load data, content, and metadata without the pain of upfront data modeling, and evolve your models alongside discovery

Reliable Performance—Transactional consistency (ACID), high availability, and government-grade security

Flexible & Scalable—Scales out on any hardware environment, adapts as data changes and new data is added

Rapid Application Development—Build out applications in days and weeks that used to take months and years

Unified Platform—Avoids the complexity and brittleness that multiple applications carry

Lower Costs—Reduces time to deployment, as well as engineering, operations, and maintenance costs

NEW GENERATION BIG DATA REQUIRES A NEW GENERATION DATABASE

Schema-agnostic Enterprise NoSQL database technology, coupled with powerful search and flexible application services—MarkLogic is the trusted platform for information applications for organizations that look to drive revenue, streamline operations, manage risk, and make the world safer.

Powerful and complete platform to speed development. Unrivaled collection of enterprise-hardened features and development tools mean that MarkLogic helps you build valuable applications fast.

Accessible and agile to ensure you see results fast. Flexible APIs let you code in the language of your choice, while integrations allow data to be available to those who need it to make better decisions.

Trusted to run the most mission-critical applications. Proven reliability, scalability, security, and over 500 deployed enterprise projects—MarkLogic is the Big Data platform public and private sector organizations depend on.

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To learn more, go to
www.marklogic.com.